

Project Idea

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1. **Air Pollution Prediction using Transfer Learning** (air pollution prediction levels in environments by using a teacher–student transfer learning framework where labeled data is limited)

Dataset

Beijing PM2.5 dataset

<https://www.kaggle.com/djhavera/beijing-pm25-data-data-set>

https://github.com/jonathanwvd/awesome-industrial-datasets/blob/master/markdown/beijing_pm2_5_data.md

[Works that used this dataset](#)

Short Overview

● Dataset Description

Number of Samples

- 43,824 instances (rows of data)

Each instance represents one hour of recorded environmental data.

Time Range

- From January 1, 2010, to December 31, 2014
- covers 5 years of hourly pollution and weather observations.

● Features in the Dataset

The dataset contains 11 features, including temporal, pollution, and meteorological variables.

1. Target Variable

- PM2.5 – concentration of fine particulate matter in the air (the value the model predicts)

2. Time Features

These describe when the observation was recorded:

- Year
- Month
- Day
- Hour

3. Meteorological Features

These influence air pollution dispersion:

- DEWP – Dew point
- TEMP – Temperature
- PRES – Atmospheric pressure
- cbwd – Combined wind direction
- lws – Cumulated wind speed
- ls – Cumulated hours of snow

- Ir – Cumulated hours of rain
- These features help the model understand how weather conditions impact pollution levels.

Dataset Characteristics

- Type: Multivariate time-series dataset
- Task: Regression (predict PM2.5 value)
- Feature types: Integer and real-valued features
- Missing values: Present (marked as "NA")
- Source: UCI Machine Learning Repository

The dataset combines air pollution measurements from the US Embassy in Beijing with meteorological data from Beijing Capital International Airport.

Summary

The Beijing PM2.5 dataset contains 43,824 hourly observations collected from 2010 to 2014. It includes 11 features, consisting of temporal variables and meteorological measurements such as temperature, pressure, wind direction, and rainfall. The target variable is PM2.5 concentration, making the problem a regression task. T

Literature Review

Nagrega et al. (2020) proposed a hybrid deep learning model combining 1D CNN and LSTM for predicting PM2.5 concentrations using only ground-based sensor data. The dataset comprised four years of hourly pollutant readings (O₃, CO, SO₂, NO₂, PM2.5, PM10) from four sites at the Port of Los Angeles, split 75/25 for training and testing. The CNN extracted local temporal features using twelve filters with a 5×1 kernel, while the LSTM captured long-term dependencies in pollutant trends. The model predicted PM2.5 levels in three-hour intervals across a twenty-four-hour window (eight steps ahead). It achieved RMSE values of 20–25% of mean site concentrations, comparable to more data-intensive models. A key strength is its reliance solely on low-cost sensor data, without requiring meteorological or satellite inputs, making it suitable for resource-limited regions. The model demonstrated robustness across varying seasonal and weather conditions. The authors argued that embedded temporal information within pollution data alone is sufficient for accurate forecasting. Future work was suggested to extend the architecture to 2D CNN-LSTM for spatial pollution mapping using satellite data. Overall, the paper contributes a lightweight, deployable framework for real-time localized air quality forecasting in urban and suburban environments. This paper establishes the deep learning baseline for pollution forecasting in data-limited, resource-constrained environments — directly motivating why transfer learning is needed when even this lightweight model struggles in regions with insufficient historical sensor data.

<https://ieeexplore.ieee.org/document/9457991/citations#citations>,

This research (2024) proposed TL-AdaBiGRU, a transfer learning-based model with a two-stage attention mechanism to predict air pollutant concentrations at new monitoring sites with limited historical data. The first stage applies temporal attention to segment and weight pollutant sequences by period, while the second stage uses a multi-head external attention mechanism to extract hidden layer key features from a Bidirectional GRU. Knowledge learned at a data-rich source domain site is then migrated to a data-scarce target site, improving prediction capability without requiring extensive retraining. The model was evaluated on atmospheric monitoring data from multiple sites in Inner Mongolia, China, outperforming baseline LSTM and GRU models in RMSE and MAE. The key contribution is addressing the cold-start problem at newly deployed monitoring stations through domain knowledge transfer. A limitation is that cross-regional generalizability across climatically diverse environments was not fully validated. This is one of the closest matches to the project: it directly implements knowledge transfer from a source domain to a data-scarce target station, which is the core mechanism of your teacher–student framework.

<https://www.nature.com/articles/s41598-024-57784-7>

This study applies transfer learning using a hybrid Graph Convolutional Network and LSTM (GCN-LSTM) to improve short-term air quality forecasting at stations where labeled data is limited or absent due to recent installation or sensor failure. The base models were trained on eleven years (2011–2022) of NO₂ and PM10 data from four AURN monitoring stations in

Wales, UK, using 42 input features including meteorological variables. Transfer learning was applied by retraining the pre-trained GCN-LSTM on only three years of data from a new target station, significantly outperforming CNN-LSTM and Transformer-based baselines. The results demonstrated that the hybrid GCN-LSTM model generalizes effectively across monitoring stations with minimal labeled data. The study contributes a practical framework for deploying early warning systems at newly installed air quality stations. A limitation is that evaluation was confined to a single country's monitoring network, restricting cross-regional applicability. It directly validates that transfer learning improves forecasting accuracy at stations where labeled data is limited or absent due to recent installation or sensor failure .

<https://link.springer.com/article/10.1007/s11869-025-01713-8>

This study proposes a deep learning framework operating under the transfer learning paradigm to overcome the limitations of sparse in-situ sensor networks for air quality prediction in smart cities. The model estimates nitrogen dioxide (NO₂) concentrations in Luxembourg by combining multimodal inputs including satellite imagery, meteorological variables, and limited ground-based sensor readings. A pre-trained deep neural network is fine-tuned on the small local dataset, substantially reducing the dependency on large labeled datasets. The transfer learning approach achieved competitive accuracy compared to models trained on full datasets, demonstrating the viability of knowledge reuse across environments. The paper contributes a deployable framework for cities lacking comprehensive monitoring infrastructure. A limitation is that the model was validated in a single European urban context, and performance in developing-country environments with different pollution profiles remains untested. It demonstrates that transfer learning overcomes sparse monitoring infrastructure in smart city environments, supporting the goal of accurate pollution prediction where labeled data collection is impractical.

<https://link.springer.com/article/10.1007/s13762-024-05722-5>

This paper proposes a transfer learning method combined with Graph Neural Networks (GNNs) to predict NO₂ concentrations at unmonitored urban locations where ground-based labeled data is sparse or unavailable. Satellite-derived NO₂ column density and meteorological inputs serve as the primary training signal, with the model fine-tuned using minimal ground-level observations from portable monitoring schemes. The GNN structure captures spatial relationships between monitoring nodes, while transfer learning enables generalization to new locations from a single initial NO₂ sample. The proposed model outperformed traditional geostatistical interpolation methods such as kriging and demonstrated strong performance at unmonitored sites. The study contributes a low-cost solution for expanding air quality monitoring coverage in data-scarce urban and suburban areas. The main limitation is that the model's performance with zero ground-truth samples at target locations was not rigorously

benchmarked. It shows that a teacher model pre-trained on satellite data can transfer knowledge to unmonitored locations with near-zero labeled data, directly paralleling teacher–student framework design.

<https://arxiv.org/html/2505.05479v1>

Hu et al. (2022) present a comprehensive survey of teacher-student learning frameworks, covering knowledge distillation as a method of transferring learned representations from a large, computationally expensive teacher model to a compact student model. The survey categorizes distillation strategies including response-based, feature-based, and relation-based knowledge transfer, discussing their trade-offs in model compression, accuracy retention, and generalization. The authors review applications across computer vision, NLP, and time-series tasks, highlighting that student models trained under teacher supervision can match or approach teacher-level performance with significantly reduced parameters. Key challenges identified include the capacity gap between teacher and student models and the risk of propagating teacher bias to the student. This work provides the theoretical foundation for teacher-student transfer learning frameworks applied in environmental forecasting tasks such as air quality prediction. A noted limitation is the survey's limited coverage of domain adaptation scenarios involving environmental or sensor-based time-series data. This paper provides the theoretical and methodological foundation for teacher–student transfer learning framework, defining the distillation mechanisms the project implements for air pollution prediction under limited labeled data.

<https://arxiv.org/pdf/2210.17332>

Ma et al. (2020) proposed a transfer learning-based stacked bidirectional long short-term memory network (TLS-BLSTM) to predict air pollutant concentrations at newly installed monitoring stations with insufficient historical data. The model was trained on data from established stations in Anhui Province, China, covering pollutants including PM_{2.5}, PM₁₀, and NO₂, and the learned weights were spatially transferred to new target stations with minimal local data. A rolling window method was applied for data preprocessing, and Spearman, Pearson, and Kendall correlation methods were used to determine the optimal input window size for each pollutant. The transferred model achieved up to 35.21% lower RMSE compared to models trained from scratch at new stations, with performance improving the less data the target station possessed. The key contribution is demonstrating that spatial transfer learning can directly substitute for data collection time at new stations, offering a practical solution to the cold-start problem in air quality monitoring networks. A limitation is that the study was confined to a single Chinese province, and cross-regional or cross-climate transferability was not evaluated. This is a foundational paper because it proves that spatial transfer learning directly solves the cold-start problem at data-scarce stations, which is the core challenge the teacher–student framework is designed to address.

<https://www.sciencedirect.com/science/article/abs/pii/S0048969719357663>

This study proposed a hybrid TL-LSTM-MHA model combining transfer learning, LSTM, and Multi-Head Attention for seasonal PM_{2.5} forecasting during Delhi's stubble-burning period (September–December) using a decade of data from 2012 to 2022. The model was pre-trained on historical source data prior to 2021 to learn temporal and pollution-related patterns, then fine-tuned on recent target data to improve generalization — implementing a teacher–student style two-phase training strategy. A hybrid feature selection method (CorrXGBoost) combining Pearson correlation and Gradient Boosting significance was applied to retain the thirteen most predictive features from meteorological and FIRECOUNT biomass burning inputs. The TL-LSTM-MHA outperformed standalone LSTM, GRU, and non-transfer baselines in RMSE and MAE during peak pollution episodes. The paper contributes a fine-tuning framework that reduces dependency on large labeled target datasets while maintaining high forecasting accuracy under extreme seasonal pollution conditions. A limitation is that validation was restricted to a single city and season, leaving cross-city and year-round generalizability untested. It directly implements the two-phase pre-train and fine-tune pipeline that mirrors teacher–student framework, demonstrating its effectiveness on a real-world pollution forecasting task with limited recent labeled data.

<https://www.nature.com/articles/s41598-025-16664-4>

This paper addressed the overfitting problem in supervised air quality prediction caused by the limited number of monitoring stations by proposing a co-training semi-supervised method called KNN-DNN, which combines K-Nearest Neighbors and a Deep Neural Network to exploit both labeled and unlabeled samples. The objective was fine-grained AQI prediction at unmonitored locations, where labeled ground-truth readings are scarce while large volumes of unlabeled spatiotemporal data exist. The co-training architecture alternates between the KNN and DNN components, with each model labeling unlabeled samples for the other in an iterative self-training loop that progressively expands the effective training set. Experimental results showed that the KNN-DNN semi-supervised approach significantly outperformed purely supervised baselines trained on the same limited labeled data, with improved generalization on validation and test sets. The key contribution is demonstrating that semi-supervised co-training is a viable and practical strategy for air quality prediction under labeled data scarcity. A limitation is that the approach was evaluated on a single dataset and city, and its scalability to diverse geographic or climatic environments was not verified. It directly addresses labeled data scarcity in air quality prediction using a semi-supervised strategy, providing strong motivation and a comparable baseline for teacher–student approach which similarly exploits limited labeled environments.

<https://mdpi.com/2073-4433/14/1/143/htm>

This paper proposed a self-supervised learning framework called Multi-Task Spatio-Temporal Network (MTSTN) for fine-grained air quality inference over graph-structured sensor networks, targeting scenarios where labeled air quality data is sparse or unavailable. The architecture consists of two tasks: a self-supervised pretext task that learns rich spatiotemporal representations from unlabeled sensor data, and a supervised downstream task that uses those representations for the core air quality inference objective. The graph structure captures spatial relationships between monitoring nodes, while the temporal component models pollutant concentration dynamics over time. MTSTN outperformed purely supervised and standard semi-supervised baselines on real-world air quality datasets, demonstrating that pretext-task pretraining on unlabeled data significantly improves downstream prediction accuracy. The work contributes a novel application of self-supervised learning to environmental sensing, filling a gap left by prior semi-supervised approaches in air quality inference. A limitation is that the pretext task design requires careful tuning, and the method has not been validated across sensors from different countries or pollution regimes. It validates that pre-training on unlabeled environmental data significantly boosts downstream prediction accuracy — directly supporting the knowledge transfer principle behind teacher–student framework.

<https://arxiv.org/pdf/2408.09526>

This paper proposed a semi-supervised domain adaptation scheme for calibrating low-cost air quality sensors in target deployment environments where labeled data is scarce, while unlabeled sensor readings are abundantly available. The method introduces a weighted min-max entropy loss that prioritizes unlabeled target-domain samples closest to the labeled ones, enabling the model to leverage unannotated field data during training without requiring costly reference measurements at every location. The framework addresses the covariate shift problem — where source and target domain data distributions differ — which is a core challenge when deploying pre-trained air quality models to new geographic environments. Experimental results showed that the proposed semi-supervised domain adaptation approach outperformed competitive supervised and standard semi-supervised baselines in R^2 score and MAE on real air quality sensor datasets. The key contribution is establishing, for the first time, a semi-supervised deep learning pipeline for sensor calibration under domain shift, directly relevant to teacher–student transfer learning scenarios in pollution monitoring. A limitation is that the method was evaluated on a constrained set of sensor types and deployment sites, and its behavior under large domain gaps between very different urban environments remains unexplored. It addresses the same core challenge as the project, transferring a model's knowledge across domains when target labeled data is scarce and directly validates domain adaptation as a solution to the distribution mismatch problem in real-world pollution monitoring.

<https://arxiv.org/pdf/2210.00521>

Methodology

This study employs a student–teacher knowledge transfer framework to achieve accurate yet computationally efficient prediction. The dataset undergoes systematic preprocessing, including missing-value imputation, categorical variable encoding, and feature normalization to ensure numerical stability and unbiased learning. The processed data is partitioned into training, validation, and test subsets to prevent information leakage and enable fair performance assessment.

A high-capacity teacher network is trained using supervised learning to capture complex, high-dimensional feature interactions. Once convergence is achieved, the teacher generates softened output distributions that encode relationships beyond hard labels. A compact student network is then optimized using a composite loss function that simultaneously minimizes classification error with respect to ground-truth labels and divergence from the teacher’s output distribution. This dual-objective learning enables effective knowledge distillation while constraining model complexity. Model evaluation is conducted on unseen test data using standard metrics to validate generalization and robustness.

Domain knowledge from atmospheric physics is incorporated through meteorological variables such as temperature, humidity, wind speed, and pressure. These features encode dispersion, transport, and accumulation mechanisms of pollutants, improving model interpretability and robustness.

Student–Teacher Transfer Learning Framework

The proposed framework consists of two neural networks:

1. Teacher Model: Trained on the source domain using abundant data and physics-informed features.
2. Student Model: Trained on limited target domain data while mimicking the teacher’s predictions.

Knowledge distillation is performed by minimizing a weighted loss combining prediction error and teacher–student discrepancy. This allows the student model to inherit generalized representations learned by the teacher.

Model Architecture:

Both teacher and student models employ deep neural networks with RNN LSTM layers. The architecture is designed to balance expressiveness and computational efficiency. Regularization techniques are used to prevent overfitting in the data-scarce setting.

Experimental Setup:

Dataset description:

The experiments utilize air quality and meteorological datasets collected from multiple regions. Source domains contain long-term historical records, while target domains simulate data-scarce scenarios by subsampling observations.

Baseline Models:

The proposed approach is compared against:

XGBoost

GRU

Random Forest

LSTM / Bidirectional LSTM

Standard Deep Neural Network

Evaluation Metrics:

Model performance is evaluated using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and coefficient of determination (R^2).

Code

StudentTeacher_Physics-Based_Transf
erLearning

[CSE427 Project: StudentTeacher_Transfer Learning.ipynb](#)

Processing Pipeline

Data Processing & Implementation Pipeline

Phase	Task	Implementation Detail	Purpose
1. Data Alignment	Structural Pivot	Used <code>pivot_table</code> to convert Kihumo's long-format parameters into feature columns.	Synchronizes the Student data schema with the Teacher (Beijing) dataset.
	Schema Mapping	Renamed Kihumo columns) to match Beijing's naming convention.	Ensures the model receives consistent feature names across different data sources.
2. Feature Engineering	Synthetic Metrics	Applied the Magnus-Tetens formula to derive <code>DEWP</code> (Dew Point) from temperature and humidity.	Recreates missing meteorological variables required by the Teacher model's input layer.
3. Temporal Cleaning	Resampling	Downsampled high-frequency sensor data to an hourly ('H') mean.	Matches the hourly resolution of the Beijing dataset and reduces sensor noise.
	Imputation	Applied <code>linear interpolation</code> followed by <code>ffill/bfill</code> .	Eliminates gaps in time-series data to prevent model training errors.

4. Transfer Validation	Statistical EDA	Generated boxplots and histograms to compare PM2.5 and TEMP distributions.	Validates that Kihumo data falls within a comparable range to Beijing for transfer learning.
5. Normalization	Cross-Dataset Scaling	Applied a <code>MinMaxScaler</code> fitted on Beijing data to the Kihumo dataset.	Preserves the global feature scale so the model interprets "high" values consistently.